

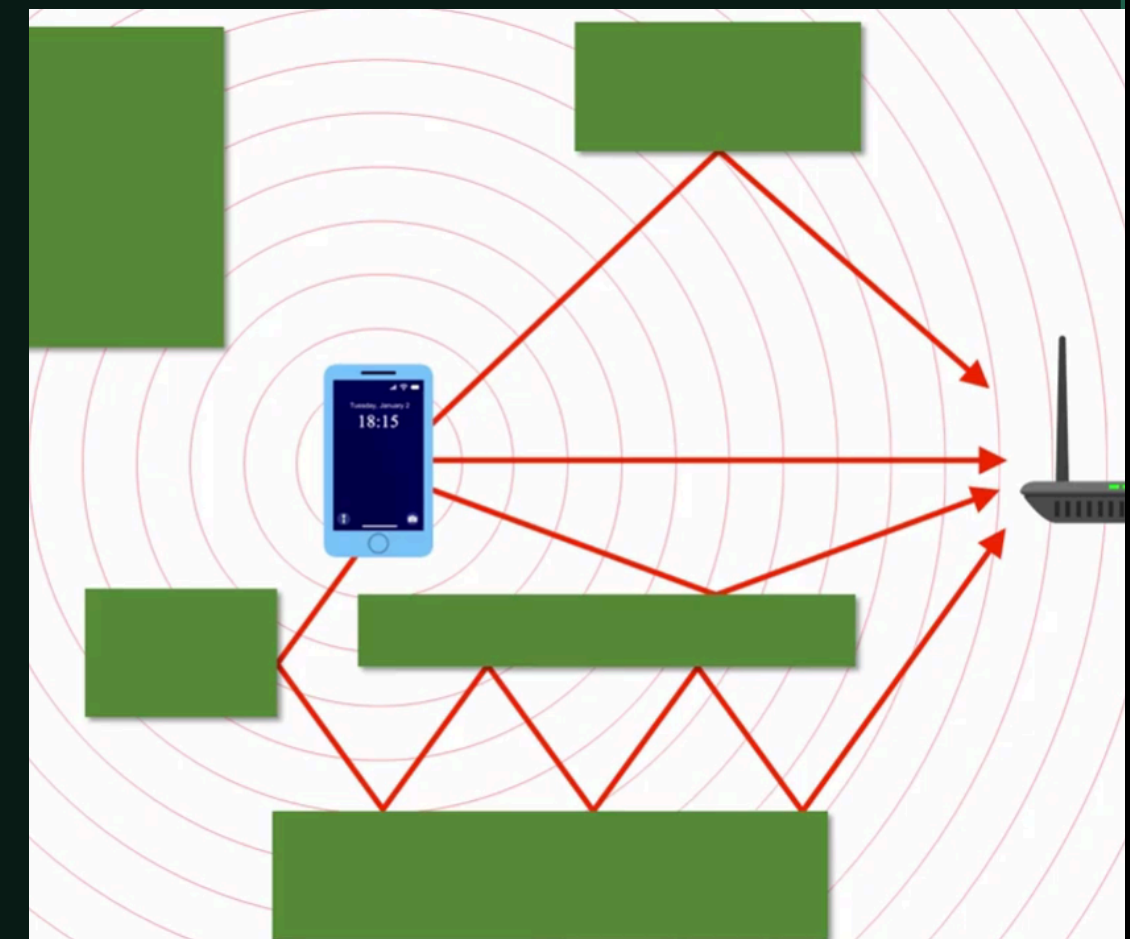
MIMO IN OFDM

Definitions, Mathematical Concepts and Simulation

Group 2:
Nguyen Quoc An (10223002)
Tran Viet Quan (10223099)
Tran Viet Hung (10223022)

1.1-INTRODUCTION

- Nowadays, with the developments in technology, the problem of fast data transmission has been increasingly relevant.
 - Inventions such as smartphones, 5G, advance Wi-Fi standards, etc. all requires a fast and reliable method of transmit and receive information.
 - The problem: Multipath Propagation, which can leads to signal delay, attenuation.
 - With increasing data density, this could cause corruption.
- Modern day telecommunication uses a combination of OFDM and MIMO system to address the problem.



1.2-DEFINITION: OFDM

- Orthogonal frequency-division multiplexing (OFDM) is a type of digital transmission used in digital modulation to encode digital (binary) data on multiple carrier frequencies
- Concept: By using orthogonal subcarrier, when one signal peaks in the frequency domain, the signal of other subcarriers reach zero.
- Benefits: more efficient bandwidth usage, higher data rate, greater resistance to interference.
- When applied to a MIMO system, we get clean signal with high data density.

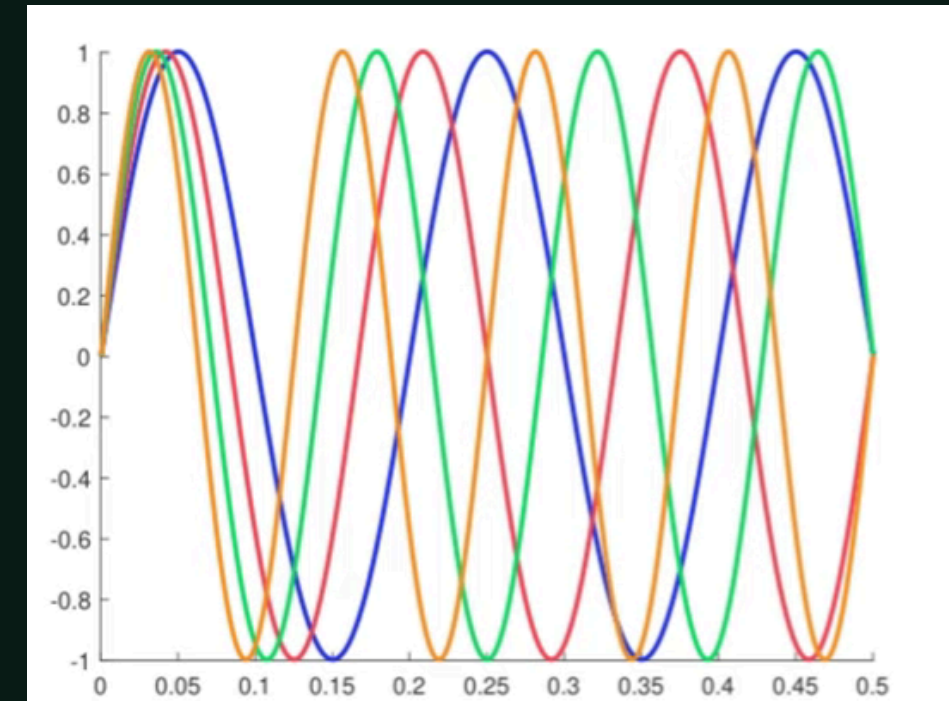


Figure: Subcarriers 5-8 - time domain signal

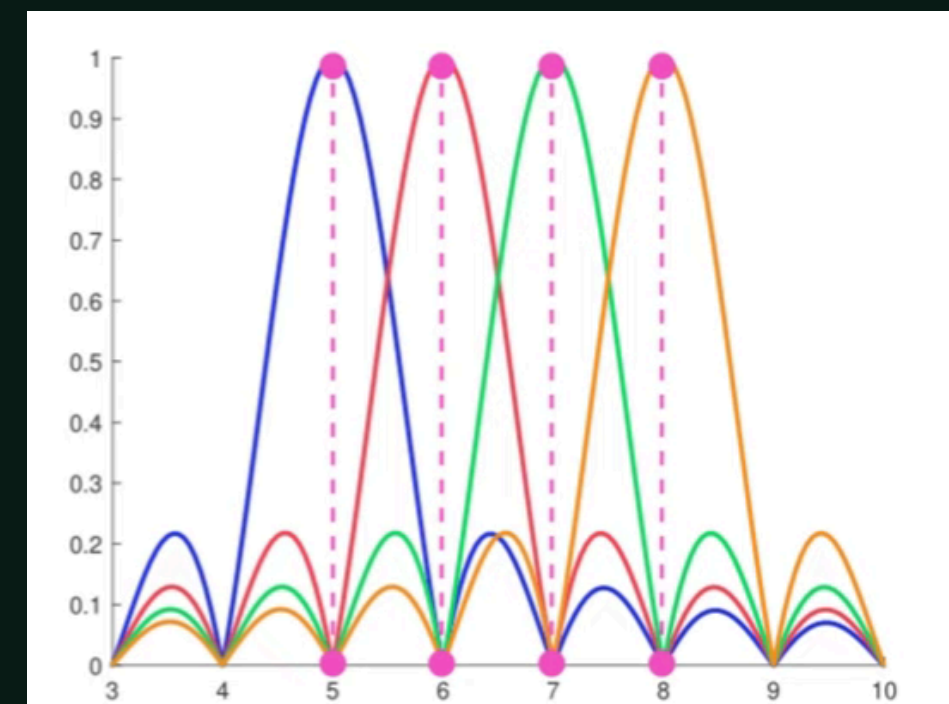
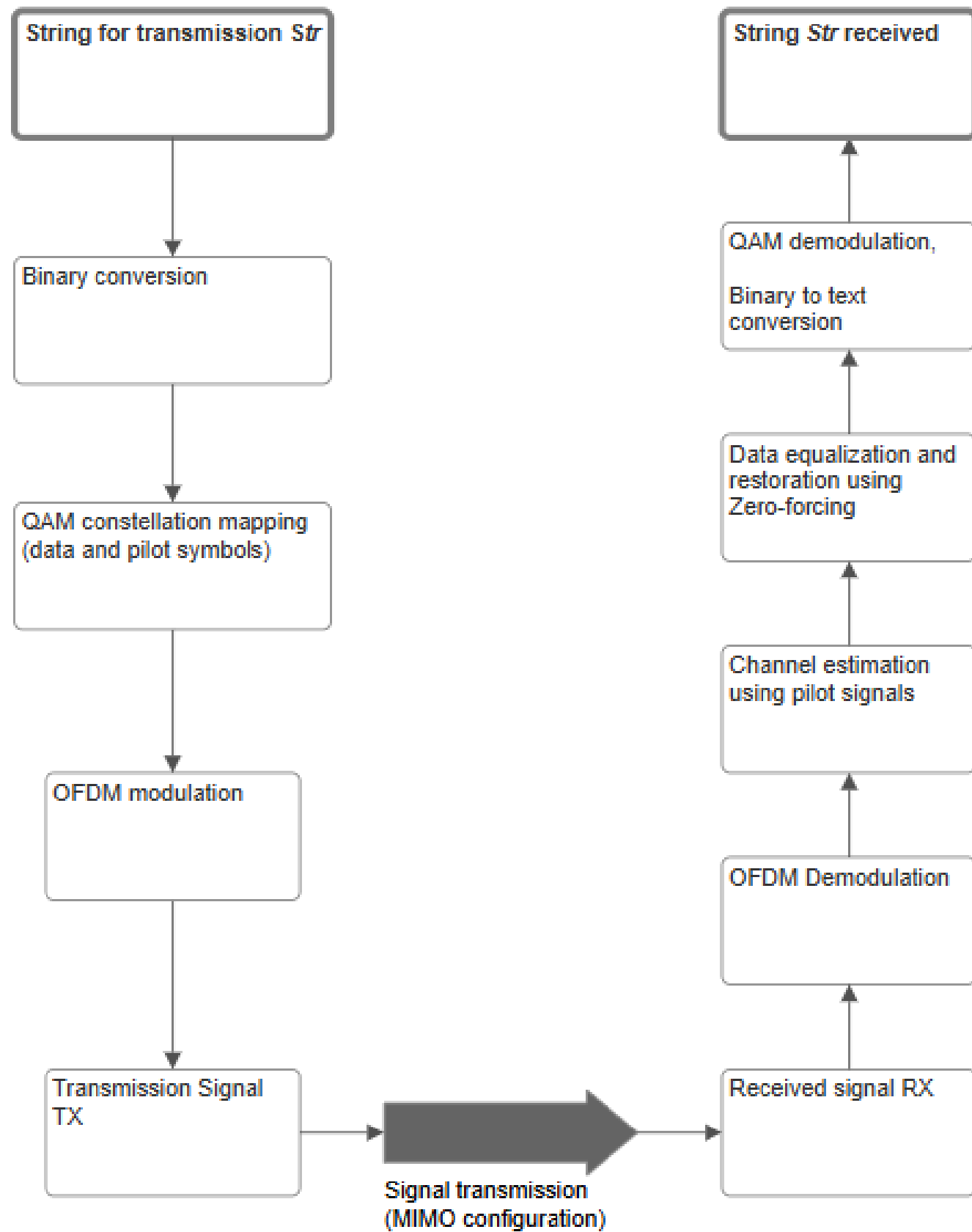


Figure: Spectrum of four OFDM subcarriers



SIMULATION OVERVIEW

- The following block diagram illustrates the procedure of a transmission using OFDM + MIMO.
- System features:
 - 2 x 2 MIMO.
 - 128 subcarriers, 20 symbols per subcarrier.
 - 16-QAM modulation for 8-bit encoding.
 - Zero-forcing equalizer.

PROGRAM IN-DEPTH: OFDM

- At this point, we want to convert our complex values to the time domain for transmission.
- Method: The inverse Fourier transform (iDFT).
- Instead of performing iDFT traditionally, we use an inverted DFT matrix with value k to ensure orthogonality.

```
% Manual OFDM
k = (0 : n_subcarriers - 1).';
n = (0 : n_subcarriers - 1);

dft_matrix = exp(-2i * pi / n_subcarriers * k * n);
idft_matrix = dft_matrix' / n_subcarriers;

%Manual fftshift (1dim)
ofdm_grid_shifted = [
tx_grid{p}(n_subcarriers / 2 + 1 : end, :); ...
tx_grid{p}(1 : n_subcarriers / 2, :)];

ofdm_signal_idft = idft_matrix * ofdm_grid_shifted;
cp_segment = ofdm_signal_idft(end-cp_length+1:end, :);
tx_signal{p} = [cp_segment; ofdm_signal_idft];
tx_signal{p} = reshape(tx_signal{p}, [1, 1]);
```

$$F^{-1} = \frac{1}{N} \begin{bmatrix} \omega^{-0 \cdot 0} & \omega^{-0 \cdot 1} & \omega^{-0 \cdot 2} & \dots & \omega^{-0 \cdot (N-1)} \\ \omega^{-1 \cdot 0} & \omega^{-1 \cdot 1} & \omega^{-1 \cdot 2} & \dots & \omega^{-1 \cdot (N-1)} \\ \omega^{-2 \cdot 0} & \omega^{-2 \cdot 1} & \omega^{-2 \cdot 2} & \dots & \omega^{-2 \cdot (N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ \omega^{-(N-1) \cdot 0} & \omega^{-(N-1) \cdot 1} & \omega^{-(N-1) \cdot 2} & \dots & \omega^{-(N-1) \cdot (N-1)} \end{bmatrix}$$

PROGRAM IN-DEPTH: THE CYCLIC PREFIX

- The cyclic prefix is a 'guardrail' to make sure that the signal does not corrupt itself during transmission (due to Intersymbol Interference)
- We append a cyclic prefix of a specific length that is a repeat of the final n -th symbols to the top at every subcarrier.

```
ofdm_signal_idft = idft_matrix * ofdm_grid_shifted;  
cp_segment = ofdm_signal_idft(end-cp_length+1:end, :);  
tx_signal{p} = [cp_segment; ofdm_signal_idft];
```

PROGRAM IN-DEPTH: ZERO-FORCING EQUALIZER

- In a MIMO system, every transmitted signal from the transmitter i to receiver j is subjected to a channel coefficient $h(j,i)$ that attenuates, delays or shifts the signal.
- The Zero-forcing Equalizer works by multiply the received signal with the inverse of the channel coefficient H .
 - Pros: Restore the transmitted signal with low complexity.
 - Cons: Can increase the AGWN.

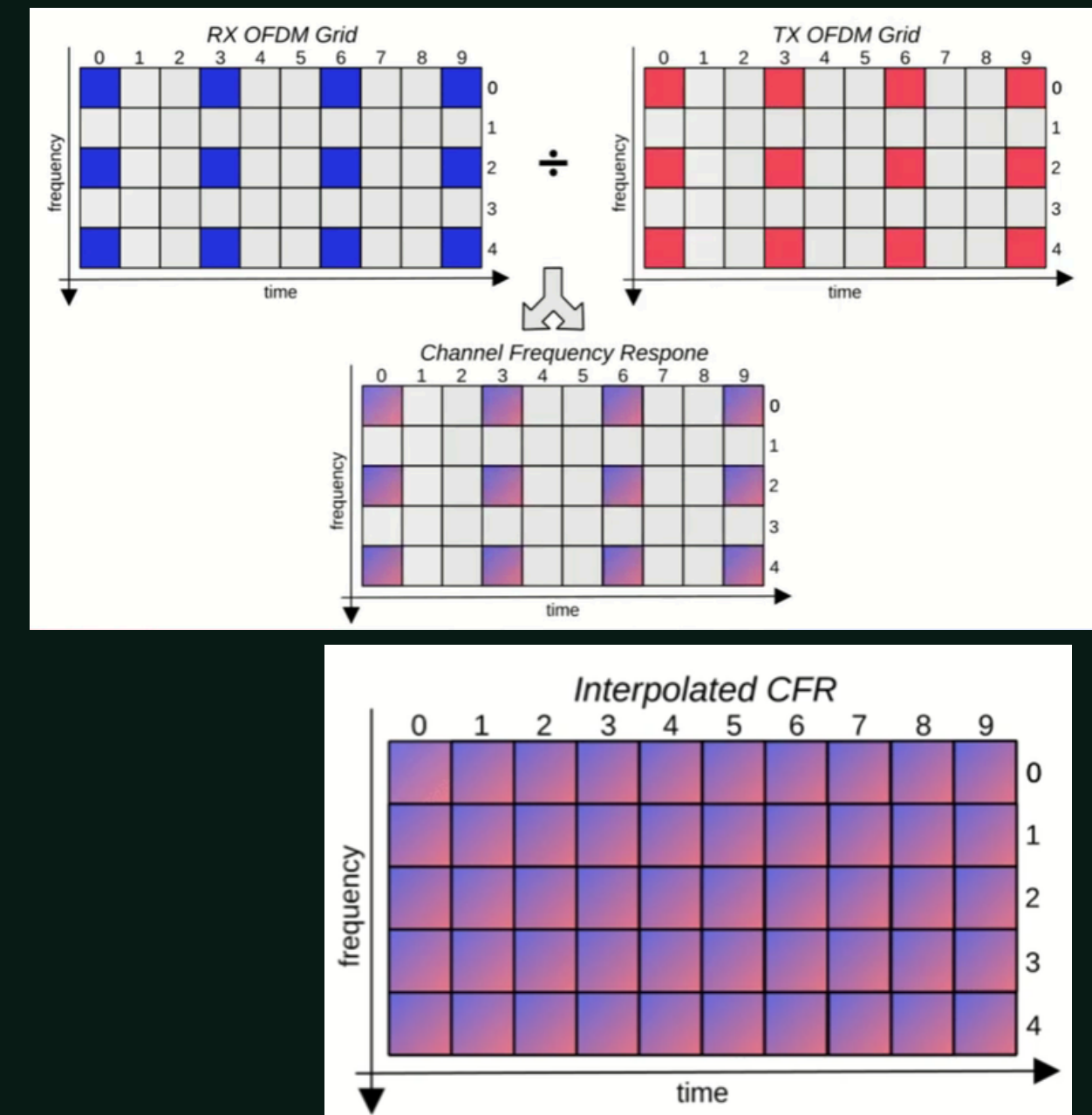
```
est_grid = zeros(n_subcarriers, n_symbols, 2);  
for k = 1 : n_subcarriers  
    for t = 1 : n_symbols  
        H = [  
            hest{1}{1}(k, t), hest{2}{1}(k, t);  
            hest{1}{2}(k, t), hest{2}{2}(k, t)  
        ];  
        y = [rx_grid{1}(k, t); rx_grid{2}(k, t)];  
        est_grid(k, t, :) = pinv(H) * y;  
    end  
end
```

But that begs the question: How can we estimate the channel response matrix?

==> We use test symbols (pilot symbols) to see how the transmission medium alters our signal.

THEORY: PILOT SIGNALS AND CHANNEL ESTIMATION

- Pick equally spaced out locations in the transmitted matrix and marked them as 'pilot locations'.
- These 'pilot locations' carries a set of values that is known to both the receiver and transmitter.
- The receiver can thus compare the arrived signal and recognizes the changes causes by the transmission medium.
- Through the proccess of 2D interpolation, the receiver can use these pilots points to estimate the channel response of every symbols for each subcarriers, thus forming the Estimated Channel Response Matrix H .



PROGRAM IN-DEPTH: PILOT SIGNALS AND CHANNEL ESTIMATION

- In this program, we install a 2-bit pilot symbol every 4 subcarriers and every 4 symbol for transmitter 1's data
- Transmitter 2's data follows the same rule, but is shifted by 2 units.
- That's around 160 pilots symbols for each transmitter.
- Both symbol sets are encoded using 4-QAM (QPSK).
- We use the built-in 2D interpolation "interp2" for matrix H estimation.

```
%% Data & Pilot Location mapping
pilot_locations{1} = false(n_subcarriers, n_symbols);
pilot_locations{1}(1 : pilot_sym_loc : end, 1 : pilot_sc_loc : end) = true;

pilot_locations{2} = false(n_subcarriers, n_symbols);
pilot_locations{2}(pilot_sc_loc / 2 + 1 : pilot_sc_loc : end, ...
    1 : pilot_sym_loc : end) = true;
```

```
%% Channel Estimation
hest = cell(n_ports, n_ports);
[X{1}, Y{1}] = meshgrid(1 : pilot_sym_loc : n_symbols, ...
    1 : pilot_sc_loc : n_subcarriers);
[X{2}, Y{2}] = meshgrid(1 : pilot_sym_loc : n_symbols, ...
    1 + pilot_sc_loc / 2 : pilot_sc_loc : n_subcarriers);
[Xq, Yq] = meshgrid(1 : n_symbols, 1 : n_subcarriers);
for p_tx = 1 : n_ports
    for p_rx = 1 : n_ports
        hest_pilots = reshape(rx_grid{p_rx}(pilot_locations{p_tx}), ...
            n_subcarriers / pilot_sc_loc, n_symbols / pilot_sym_loc) ...
            ./ pilot_states{p_tx};
        hest{p_tx}{p_rx} = interp2(X{p_tx}, Y{p_tx}, hest_pilots, ...
            Xq, Yq, "spline");
    end
end
```

RUN THE SIMULATION

```
mimoChan = comm.MIMOChannel('SampleRate', fs, ...  
    'PathDelays', [0 1.5e-6], ...  
    'AveragePathGains', [0 -5], ...  
    'MaximumDopplerShift', 20, ...  
    'SpatialCorrelationSpecification', 'None', ...  
    'NumTransmitAntennas', 2, ...  
    'NumReceiveAntennas', 2, ...  
    'RandomStream', 'mt19937ar with seed', ...  
    'Seed', 12);
```

- We use the function “MIMOChannel” from Matlab’s communication toolbox to simulate a MIMO transmission medium.
- For the initial run, the parameters are set as follows:
 - *Sampling frequency: 1920×10^3 (Hz)*
 - *Delay: 1.5×10^{-6} (seconds)*
 - *Attenuation: 5 (dB)*
 - *Doppler Shift: 20 (Hz)*
 - *No spatial correlation.*
 - *2 x 2 MIMO simulation*

We observe the steps of transmission with the string *"I love Mobile Communication"* from TX1.

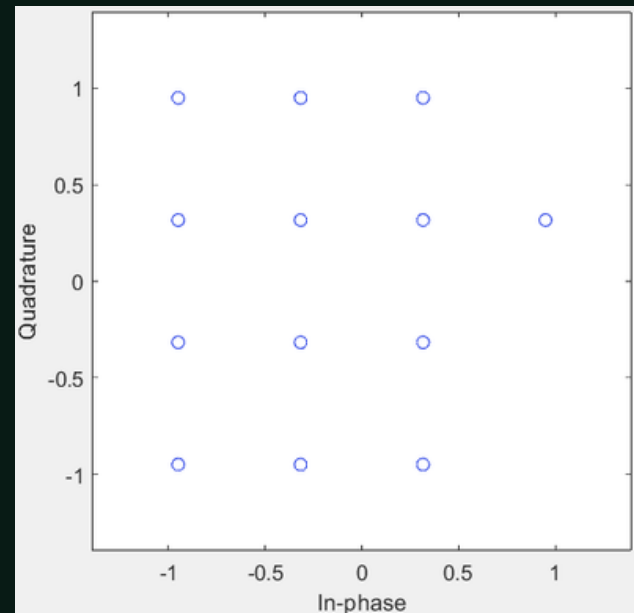


Figure 1: QAM modulation of the string before transmission.

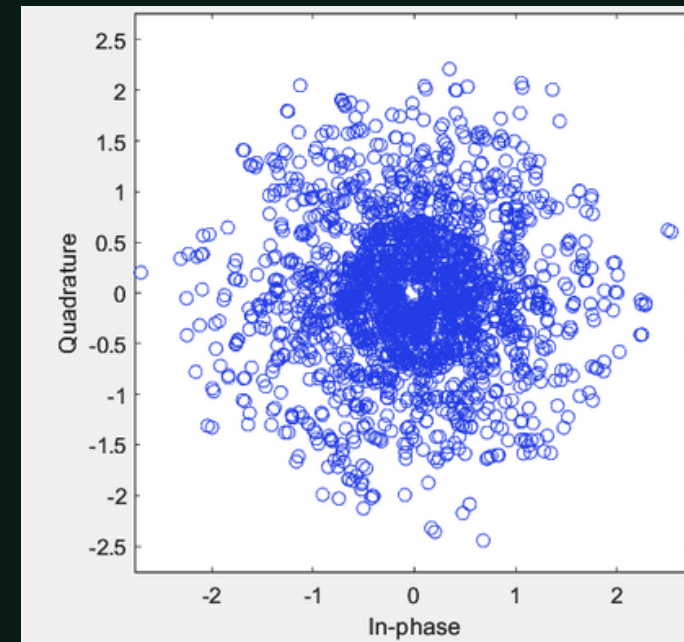


Figure 2: Received constellation

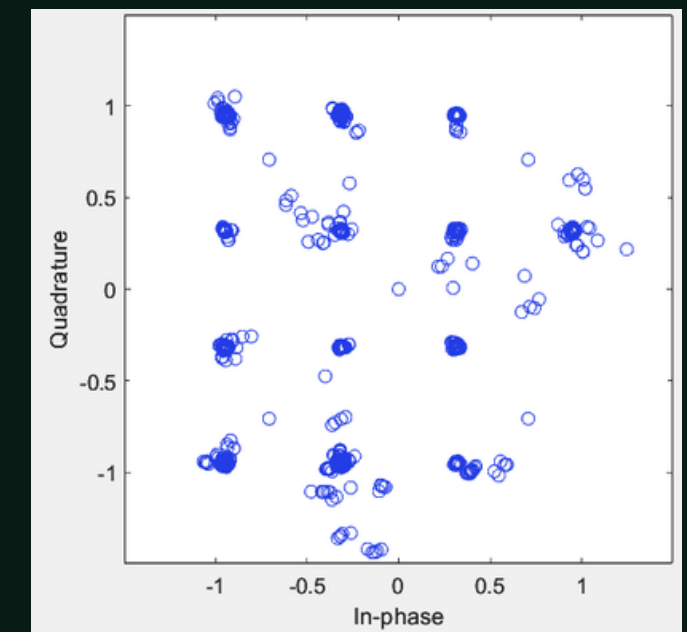


Figure 3: Received constellation after channel estimation

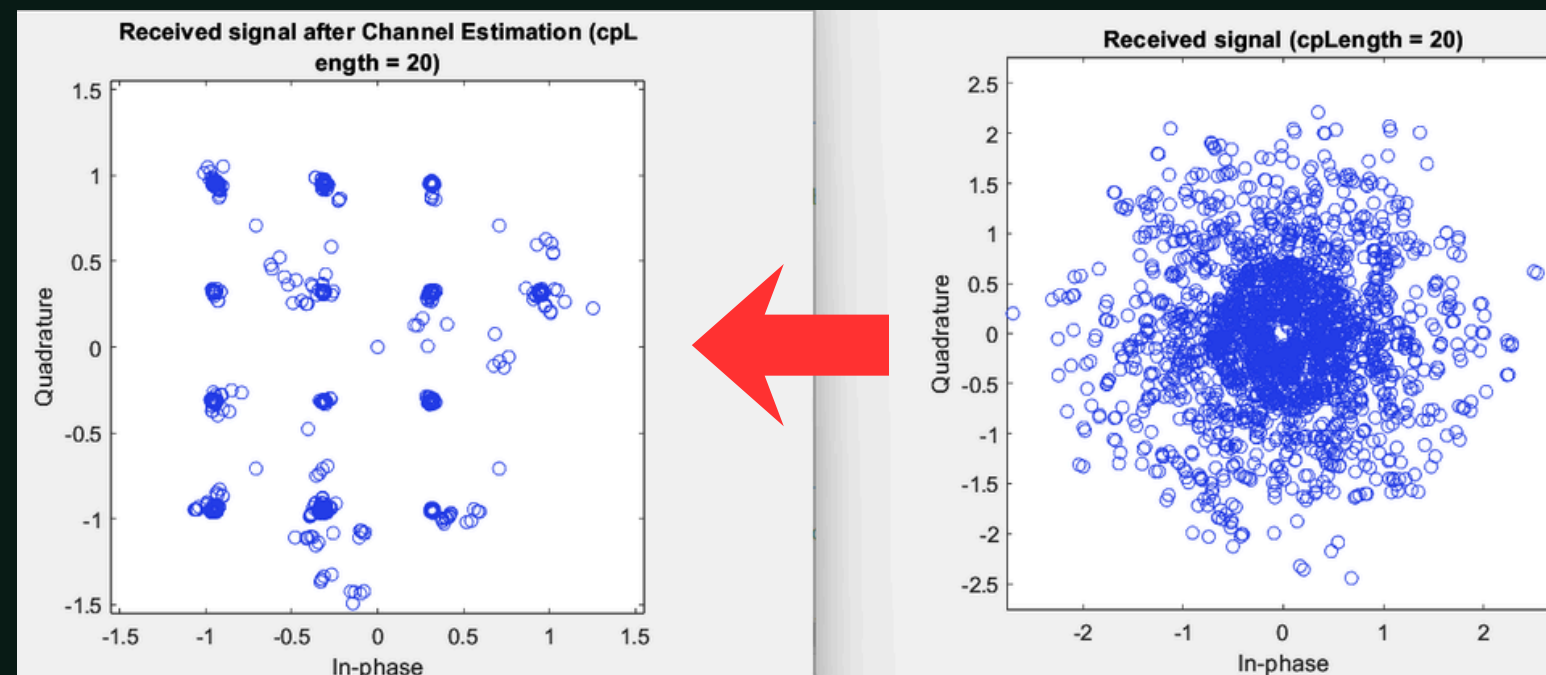
```
Port 1
BER: 0.0061384
Data: I love Mobile Communication I lmfe Mobile Communication I love Mobile Communication I love Mobilaë Communication I love Mobile Communication I love Mobile Commuéication I lov
```

Figure 4: The Bit Error Rate (BER) and the output string (repeated)

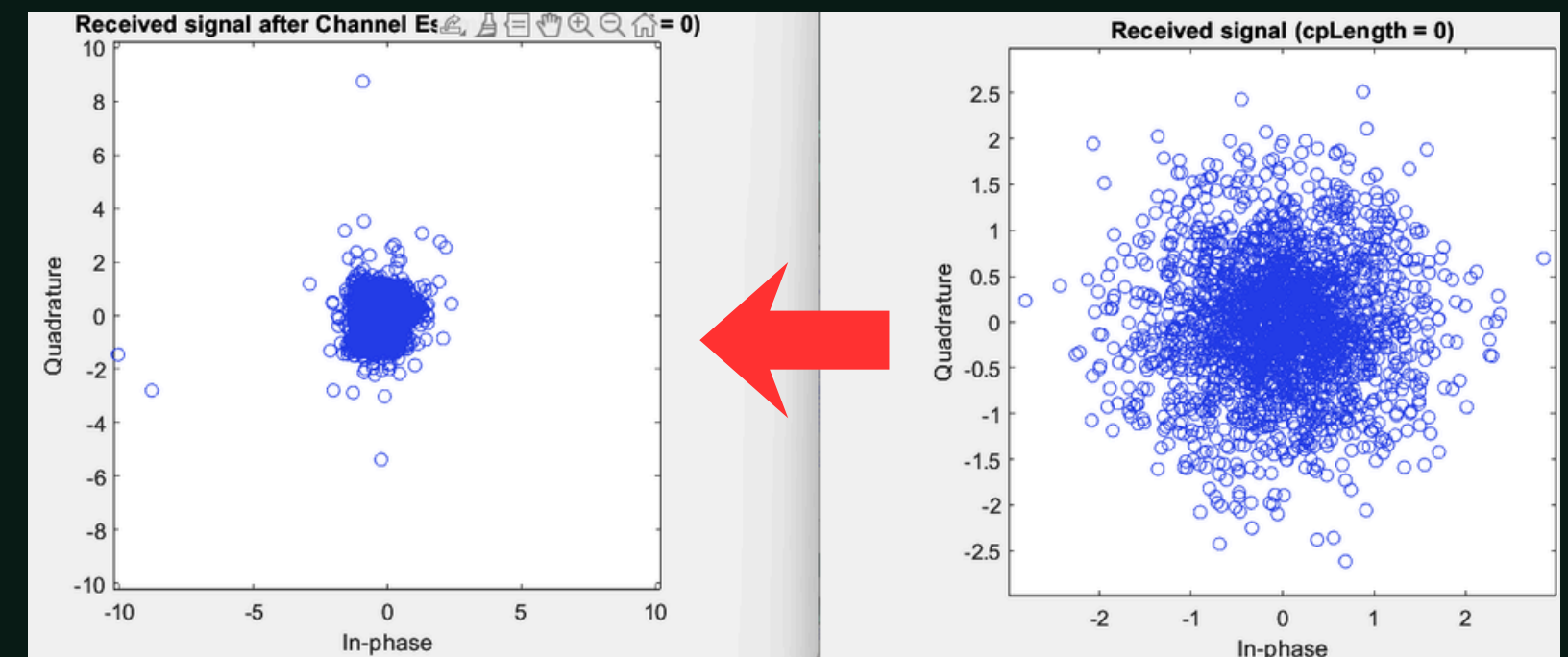
OBSERVATION UNDER ALTERNATIVE SCENARIOS

Scenario 1: No Cyclic prefix

- Initially, the cyclic prefix length (CPL) was 20 (samples).
- From the diagrams, we can see the equalizer was unable to separate the points for reconstruct when the cyclic prefix was removed.



Cyclic prefix length = 20



Cyclic prefix length = 0

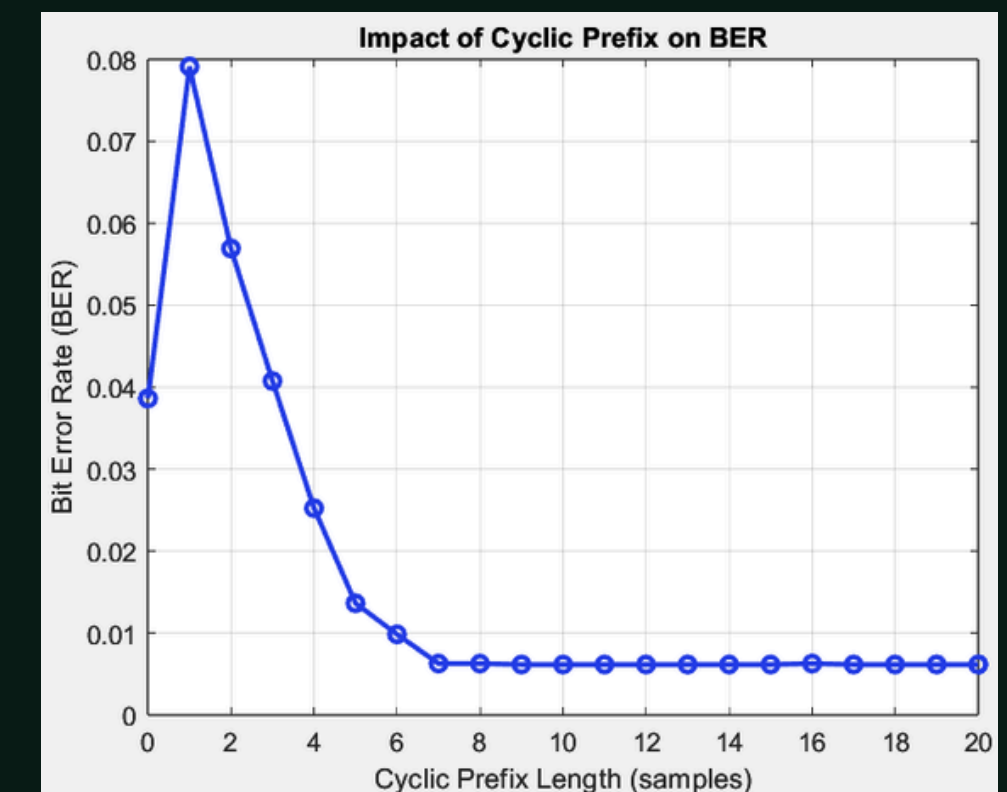
OBSERVATION UNDER ALTERNATIVE SCENARIOS

Scenario 1: No Cyclic prefix

- Plotting BER with respect to the CPL from 0 to 20:
- Conclusion: It is crucial to find the appropriate length for the CPL:
 - To short → A big increase in data corruption
 - To long → Reduced data bandwidth with negligible gain.

```
Cyclic prefix length = 0
Port 1
BER: 0.06875
Data: I love Iobm|e Boémunicat(on I lofe Mobilm Comiunic#4ion$I$hove Mob),e CommunmgcukonaI$hove Mobi,e Co}munhca
```

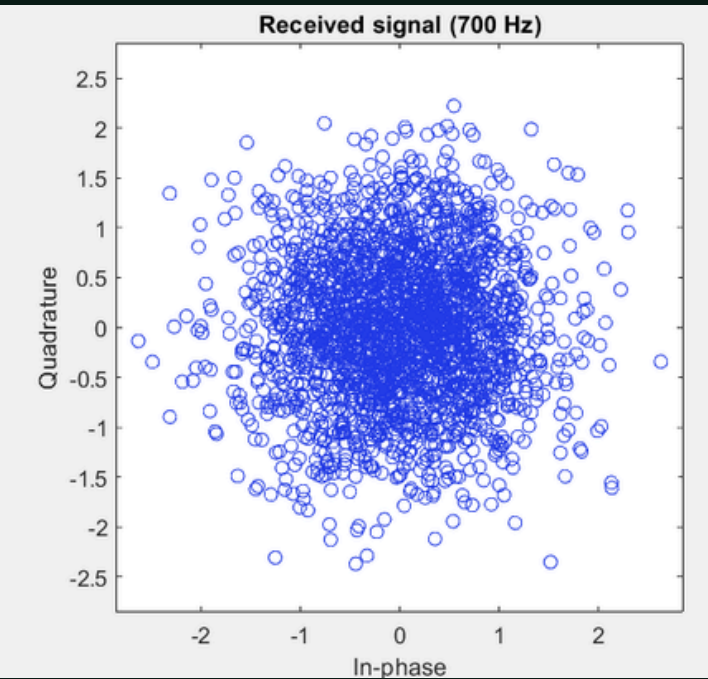
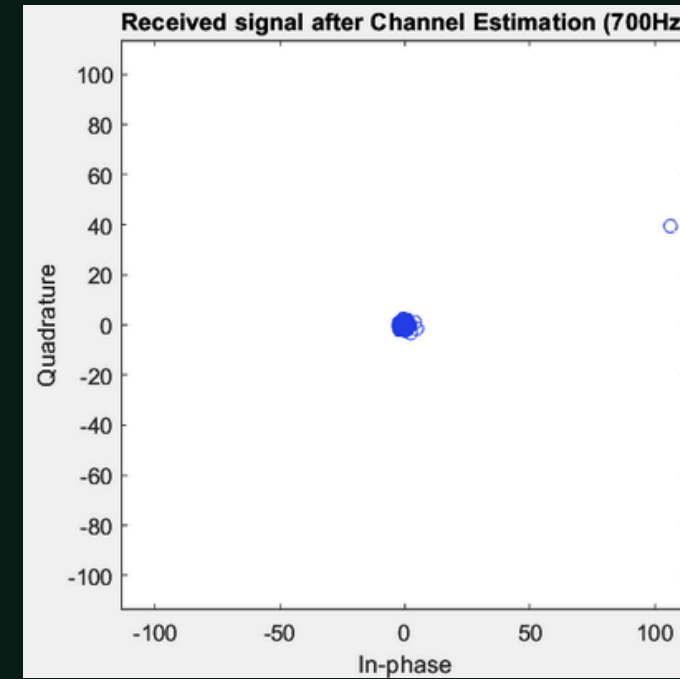
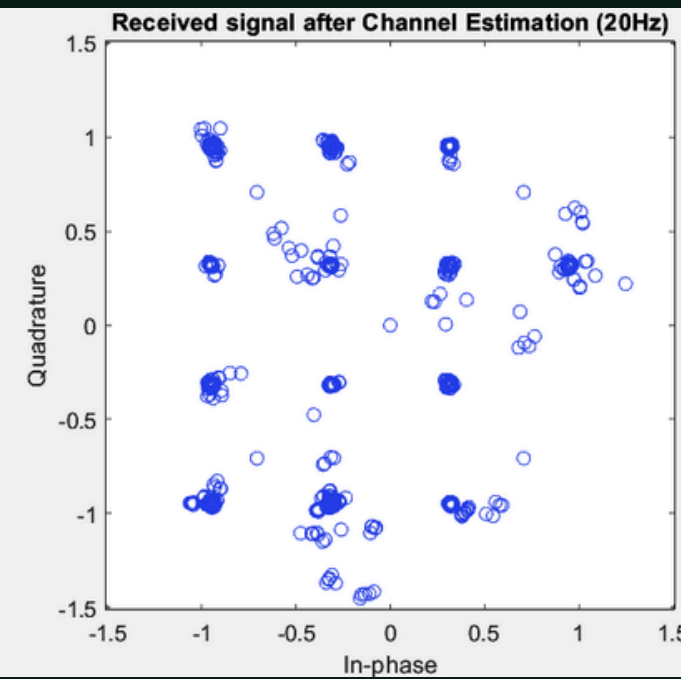
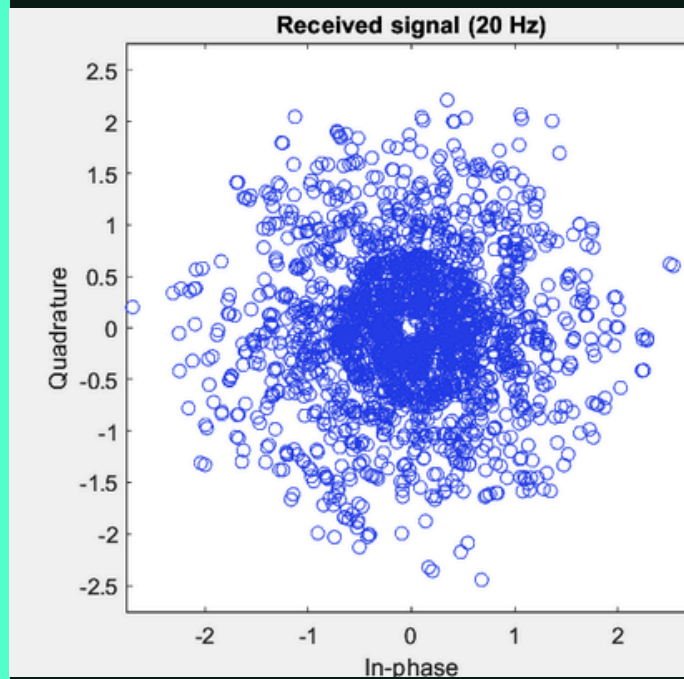
```
Cyclic prefix length = 20
Port 1
BER: 0.0061384
Data: I love Mobile Communication I lmfe Mobile Communication I love Mobile Communication I love Mobiåe Communic
```



OBSERVATION UNDER ALTERNATIVE SCENARIOS

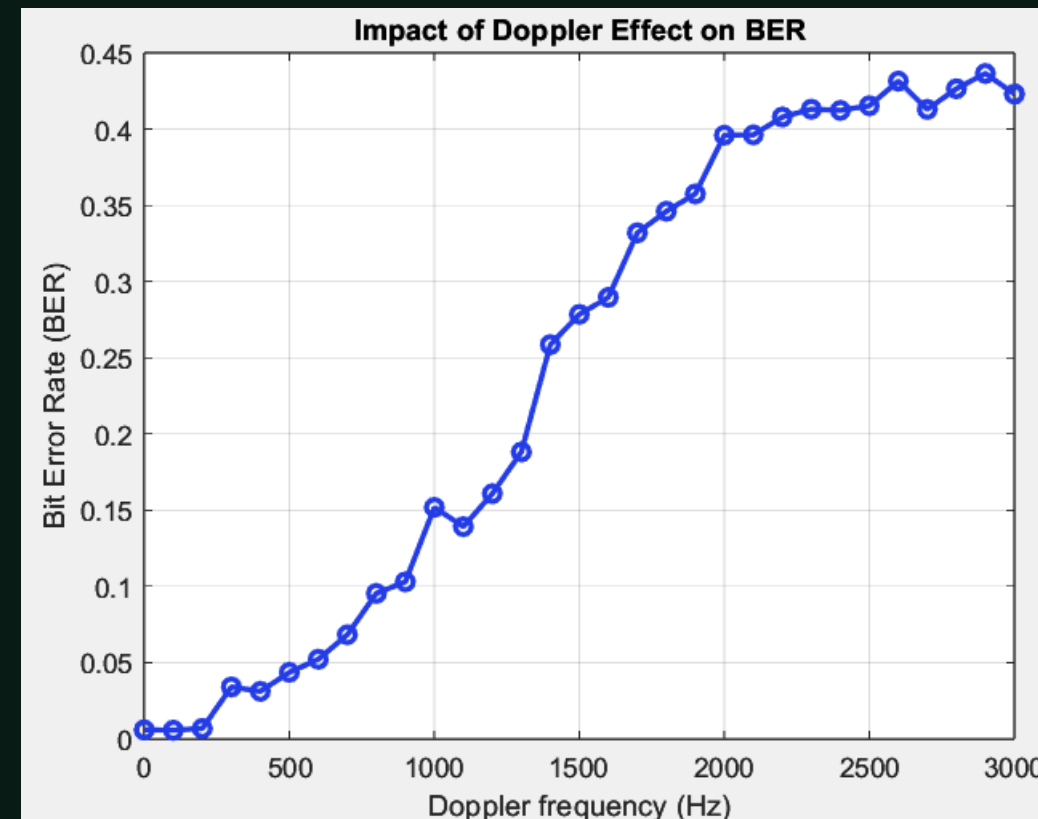
Scenario 2: Changing the Doppler Frequency

- When the receiver is moving toward or away from the transmitter, communication suffers from the Doppler effect.
- Compare Doppler frequency of 20Hz and 700Hz.
→ Failed to reconstruct the signal.



OBSERVATION UNDER ALTERNATIVE SCENARIOS

Scenario 2: Changing the Doppler Frequency



- Conclusion: The faster the receiver is moving with respect to the transmitter, the bigger the Doppler effect, and the signal gets more shifted, affecting the final result.

**THANK YOU
FOR LISTENING**